Look Before Acting: Enhancing Vision Foundation Representations for Vision-Language-Action Models

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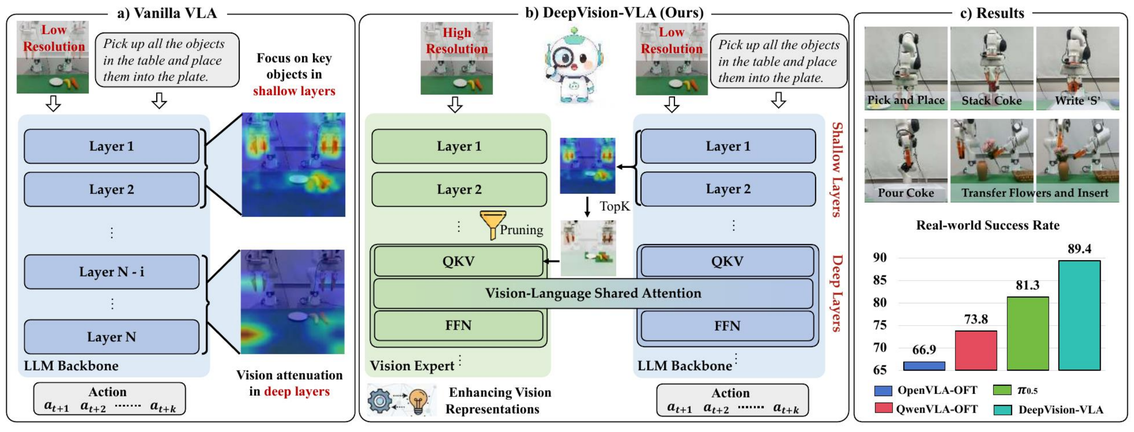
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https://deepvision-vla.github.io/
https://arxiv.org/abs/2603.15618

Overview: DeepVision-VLA



A robot understands "put the fruit on the plate", yet grasps beside it. What gets lost inside the VLA?

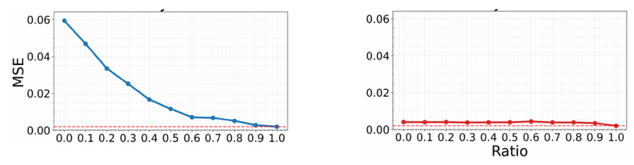
The image enters early, but task-relevant visual evidence can become underused as action prediction goes deeper.

Our response: refresh deep-layer visual evidence, guided by shallow action attention.

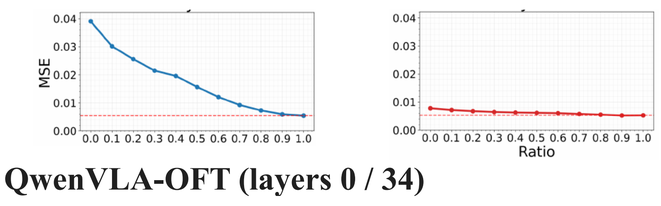
Diagnosis: why vision weakens

ROI masking: shallow / deep

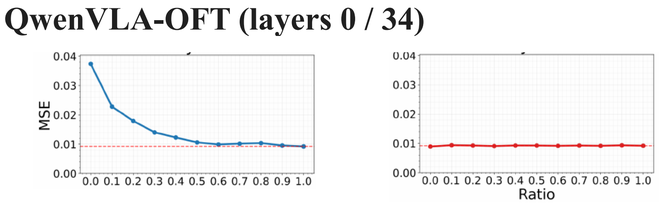
OpenVLA (layers 0 / 30)



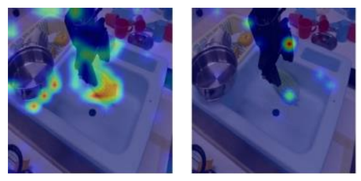
π_0 (layers 0 / 16)

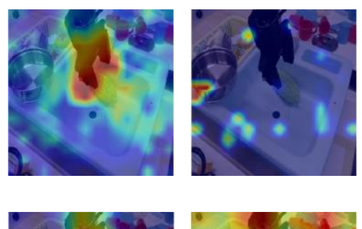


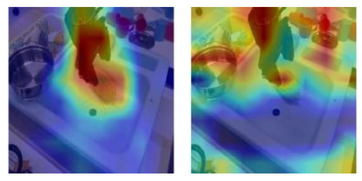
QwenVLA-OFT (layers 0 / 34)



Contribution: early / late



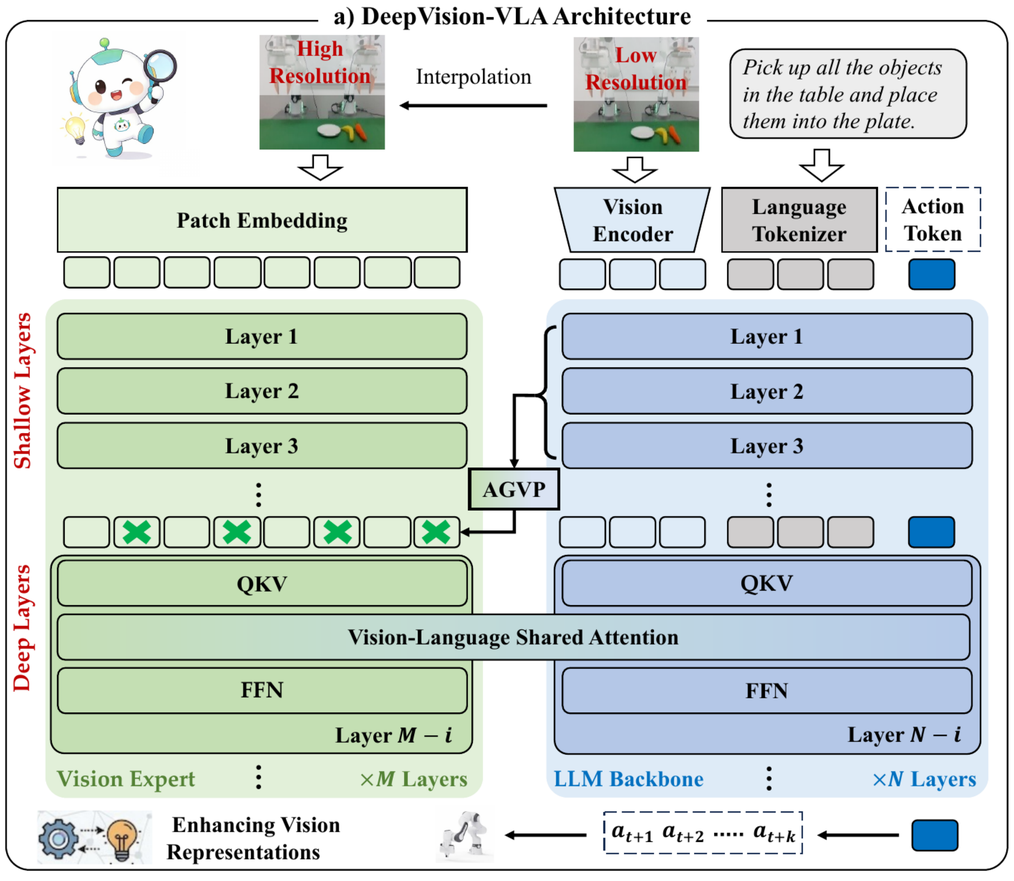


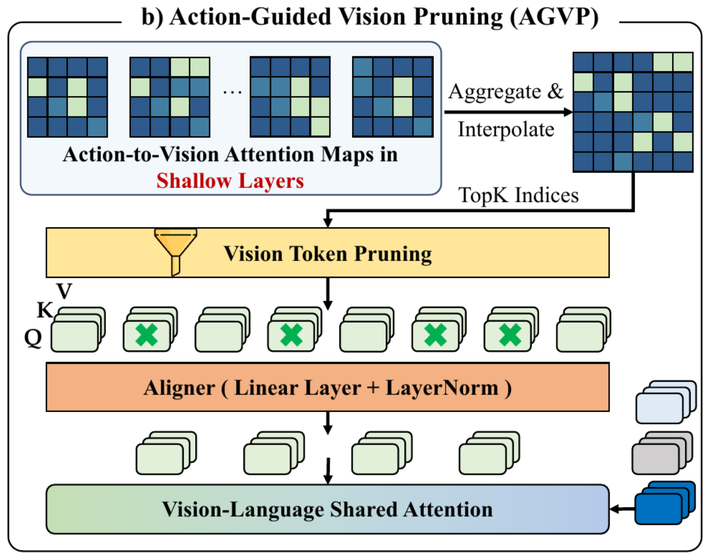


In three VLA paradigms on 1,500 BridgeV2 trajectories, ROI masking affects shallow layers more; deep-layer contribution maps lose task focus.

Insight: reinforce visual evidence in the deep layers.

Method: VL-MoT + Action-Guided Visual Pruning





1. Multi-level visual injection

The last 16 expert and VLA layers share attention.

2. Action-guided selection

Use shallow action-to-vision attention to rank expert tokens. Keep the most relevant 50%.

3. Parallel action decoding

Action tokens attend bidirectionally while prompts retain causal attention. A two-layer MLP maps the final action embeddings to continuous actions.

Shared attention refreshes visual grounding; AGVP controls how many expert tokens enter the deep layers.

Results: simulation and real robots

Simulation: RLBench

10 tasks: articulated-object control, placement, retrieval, sweeping and watering.

Model	SR (%)	Model	SR (%)
OpenVLA	40	$\pi_{0.5}$	65
SpatialVLA	46	HybridVLA	74
CogACT	61	QwenVLA-OFT	69
CoT-VLA	66	**DeepVision-VLA**	**83**

Real world: precise, multi-stage manipulation

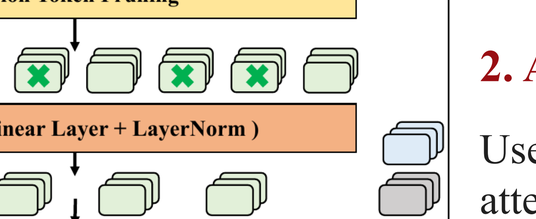
Stacking, writing, pick-and-place, pouring and bimanual transfer.

Model	SR (%)
OpenVLA-OFT	66.9
QwenVLA-OFT	73.8
$\pi_{0.5}$	81.3
DeepVision-VLA	**89.4**

The real-robot gains suggest that deeper visual grounding can support more reliable spatial interaction.

Ablations: controlled design insights

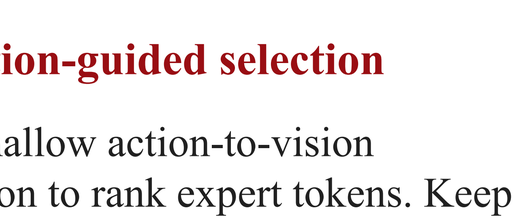
A. How to fuse vision?



Deep fusion performs best.

Deep access may refresh visual evidence where it becomes underused.

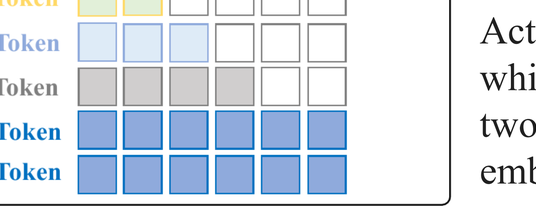
B. Which expert features?



Late DINOv3 features work best.

Late features may better capture object structure and spatial relations.

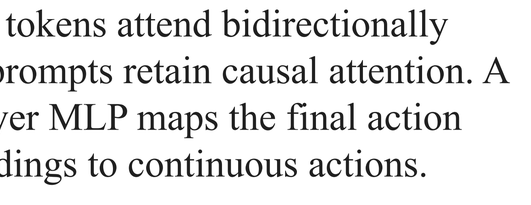
C. What guides pruning?



Action guidance works best.

Action cues may better localize the arm–object interaction required for control.

D. Which attention layers?



Averaging beats single layers.

Averaging can suppress noisy per-layer responses and stabilize token selection.

Together, the ablations favor deep-layer visual injection with late expert features and action-aware, multi-layer token selection.